Koneru Lakshmaiah Education Foundation
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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Dr Sukanya Ledalla
Student Name	Bhavana Kondakrindi
Roll Number	2520030082
Branch	CSE
Course Outcome	CO1 – Augmented AVL Tree

Case Study Topic: Real-Time Dynamic Median Rating Estimation Using an Augmented AVL Tree

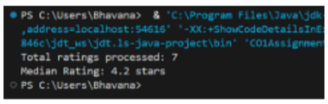
Work Done Summary:

In this case study, we implemented an Augmented AVL Tree to efficiently calculate the median of a dynamically updating stream of user ratings. By augmenting each tree node with a subtree size field, the system computes the exact median in O(\log n) time complexity via a structural descent algorithm, balancing itself automatically to prevent performance degradation.

Implementation Details:

1. Created an augmented AVL tree node structure containing rating data, height, and subtree size.
2. Implemented dynamic insertion with auto-balancing rotations to maintain an O(\log n) height.
3. Added automatic maintenance overhead to recalculate subtree sizes during structural modifications.
4. Implemented a tree descent algorithm to locate the k-th element based on rank criteria.
5. Added exact calculations to handle median retrieval for both odd and even dataset dimensions.

Result: Screenshots/output



GitHub Link:
https://github.com/bhavanakondakrindi/DSA-2-PROJECT/blob/main/CO1Assignment.java

Student Signature	Faculty Signature